Week 6

27 March 2026 20:03

Week 6 → GAN's → Generative Adversarial Network

NN

Models

Discriminative

X → Model → P(Y)

Input

Label

Generative Model

Generates data

X → random noise

P(X) → GAN

GAN → Neural Network

- Generator (G) → generate fake images
- Discriminator (D) → Detect fake images
  - → Distinguish b/w real image and fake image

Obj :- Detect the final real image very well

Training Cycle

→ Dataset
- Real images
- Random images (Random Noise) 50 → 50

→ Train the Discriminator

- Samples of Real images from Dataset
- Give it fake images

Train my generator

Convert random noise to a fake apple image

Fake Image

When the fake img is fairly close to real.

→ Complete the Discriminator training

When it distinguishes perfectly between real & fake

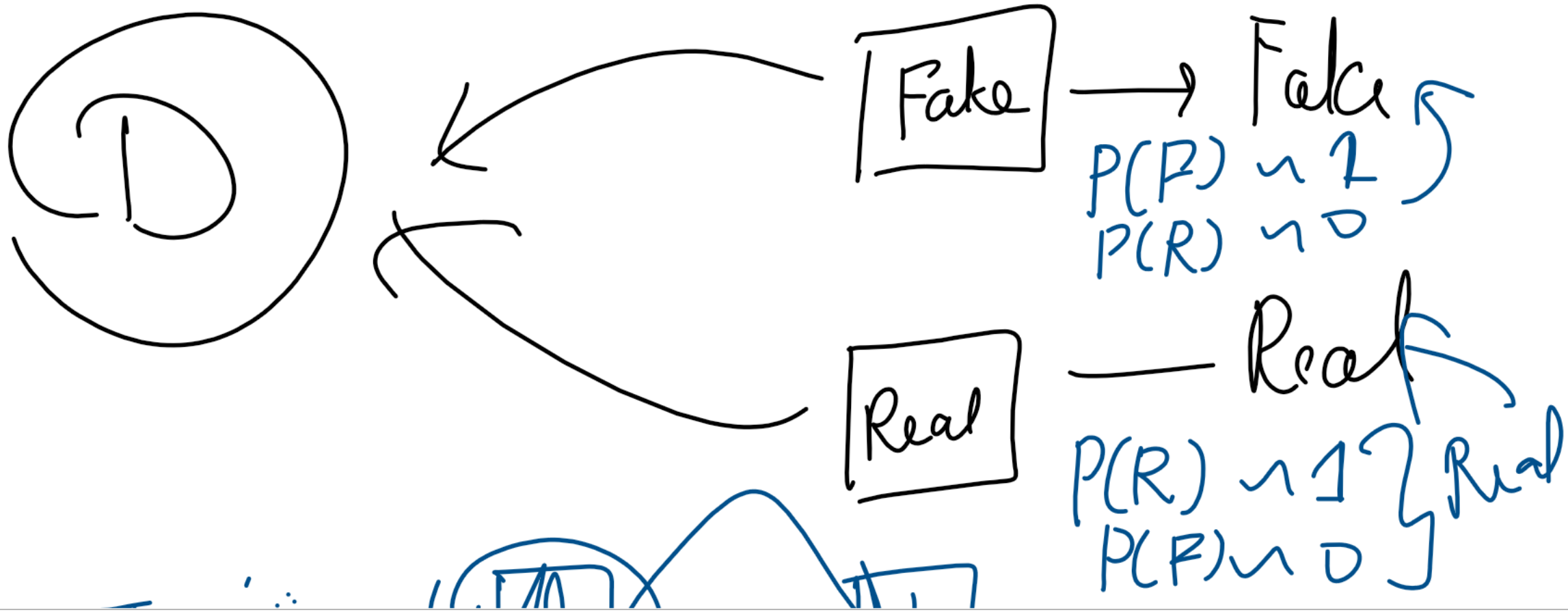
P(R)

$P(\bar{R}) \sim 1$

$P(F) \sim 0$

$P(R) \sim 0$

$P(F) \sim 1$



Input

O/P

Fake image

1	0
0	1

Fake apple image

Discriminator

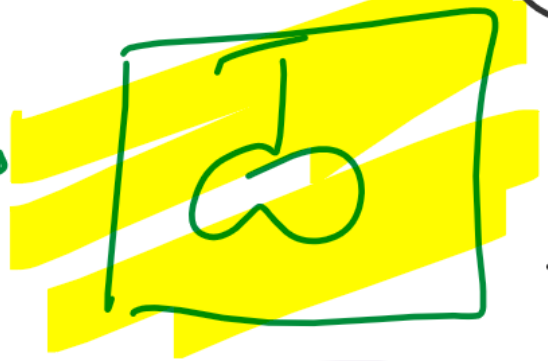
Input → Real image → Data

Input → Fake image → Generator

P(R)
P(F)

Apple

Dataset → Real image of apple →



Dataset → Random →

Random Noise (Random distorted Image)

1

- Normal Dist
- Uniform Dist

Uniform(1,2)

Uniform(a,b) → $\frac{a+b}{2}$

1.5	1.5
1.5	1.5

Gaussian Noise

Normal

0.7	

② Take a healthy image and corrupt it

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1.5 & 1.5 \\ 1.5 & 1.5 \end{bmatrix} = \begin{bmatrix} 2.5 & 1.5 \\ 1.5 & 2.5 \end{bmatrix}$$

Losses 1) Real

$D(G(x))$

① Discriminator

$-[\log[Dx] + (\log(1 - D(G(z))))$

Real : $\log$ High

② Generator Loss

$-\log D(G(z))$

$-\log(d(x)) \sim \boxed{-\ln(\hat{y})}$

cross entropy

$-\log(d(x))$ Maximize the prob of correctly predict Real and fake images

Generator Loss

$$-\log\left(D(G(z))\right)$$

low

D(scrimanctiss probability of being real

~~Generated~~ Fake Image

D(Fake)

log(something low)

$-\log$(something high)

Loss :- $\log(D(G(z)))$

$G(z)$ → Fake Image

$D(\cdot)$ → Probability of image being real

Model is good

Model is bad

- - - - Generator is Good - - - -

Discriminator is good

$D(G(z)) \sim 0$ (low)

$-(\log(D(G(z))))$

low ∴ loss will be low

Backprop

Training

Discriminator is bad

$D(G(z))$ = high ~ 1

$-\log(D(G(z)))$

high → loss will be high

$D(G(z)) = 0.1 \rightarrow \text{loss} = +1$

$D(G(z)) = 0.01 \rightarrow \text{loss} = +2$

<u>Loss</u> $\rightarrow$ $\underline{-\log(D(G(z)))}$

Loss will increase when Discriminator predicts the Generated image to be real $\rightarrow P(R) \sim 1$

Loss will reduce when D predicts the Generated image to be not real $\rightarrow P(R) \sim 0$

Issues of GAN Training

Issues of GAN Training

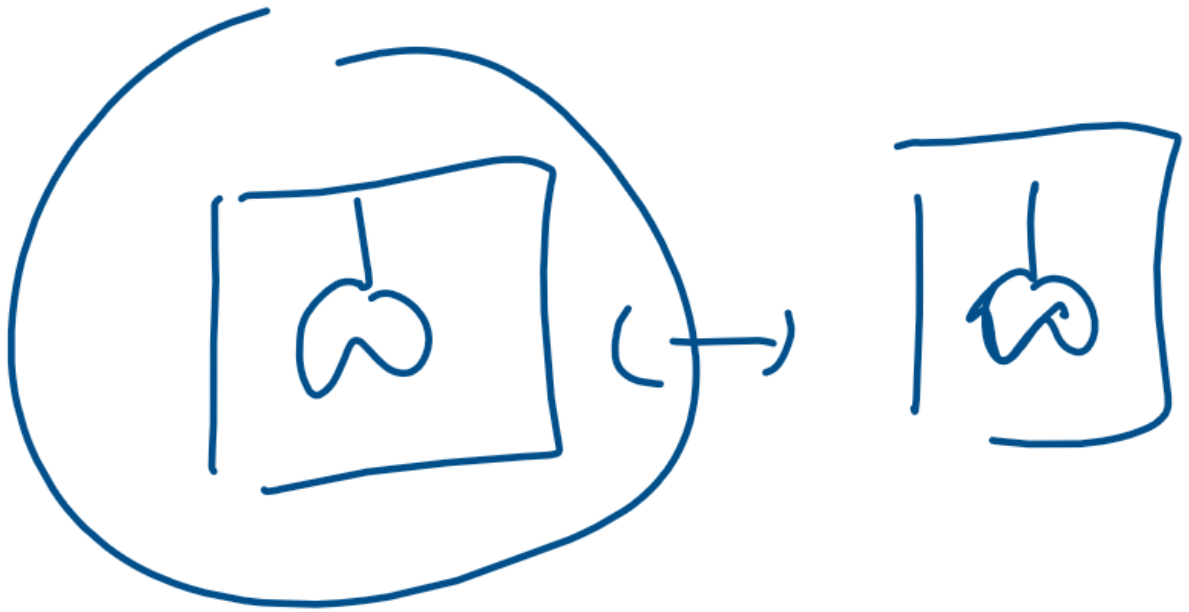
- Mode Collapse
- Vanishing Gradient

$$W_t = W_{t-1} - \eta \nabla_{W_{t-1}} \mathcal{L}$$

$$W_t \approx W_{t-1}$$

- Training Instability

Mode Collapse

↳ Generator → Generated Images are like copies ⟷ Very similar

→ lack diversity